

Parallel practice set (new numbers, same problem types)

1. $H(t)$ is the height (cm) of a snow pile, $t =$ days since it started. Explain $H'(9) = 2.4$.

$H'(t)$ = instantaneous rate of change of height (cm per day)

$$H'(9) = 2.4 \Rightarrow \text{at } t = 9$$

On day 9, the height is increasing at a rate of 2.4 cm per day.

2. For what k and m is g continuous and differentiable? $g(x) = \begin{cases} 4x + k & x \leq 1 \\ x^3 - mx + 2 & x > 1 \end{cases}$

Continuous at $x = 1$: $4(1) + k = 1^3 - m(1) + 2$

$$4 + k = 3 - m \Rightarrow k = -1 - m$$

Differentiable at $x = 1$: $g'(x) = \begin{cases} 4 & x < 1 \\ 3x^2 - m & x > 1 \end{cases}$

$$4 = 3(1)^2 - m \Rightarrow m = -1$$

$$k = -1 - (-1) = 0$$

Check: $4(1) + 0 = 4, \quad 1 + 1 + 2 = 4 \quad \checkmark$

$k = 0, \quad m = -1$

3. Which limit gives the derivative of $f(x) = 4x - x^2 + 3$?

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$f(x+h) = 4(x+h) - (x+h)^2 + 3$$

$$f(x) = 4x - x^2 + 3 \quad (\text{subtract the whole thing})$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{4(x+h) - (x+h)^2 + 3 - (4x - x^2 + 3)}{h}$$

4. Find the derivative at the point (4 decimals): $f(x) = \sec^2(2x)$ at $x = 1$.

Calculator: math $\rightarrow$ nDeriv

$$\frac{d}{dx} \left(\frac{1}{(\cos(2x))^2} \right) \Big|_{x=1} \approx -50.4692$$

Check by hand: $f'(x) = 4 \sec^2(2x) \tan(2x) \Rightarrow f'(1) \approx -50.4692$

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5. Find the derivative at the point (4 decimals): $f(x) = \frac{\sqrt{3e^x + 4}}{x}$ at $x = -1$.

Calculator: nDeriv $(\sqrt{3e^x + 4}/x, \quad x, \quad -1)$

Check by hand (quotient rule): $u = \sqrt{3e^x + 4}, \quad u' = \frac{3e^x}{2\sqrt{3e^x + 4}}$

$$f'(x) = \frac{x\,u' - u}{x^2} \Rightarrow f'(-1) \approx -2.5034$$

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6. Average rate of change of $S(x)$ on $8 \leq x \leq 13$.

x	5	8	10	13	15
$S(x)$	20	26	31	40	47

$$\begin{aligned} \text{AROC} &= \frac{S(13) - S(8)}{13 - 8} \\ &= \frac{40 - 26}{5} = \frac{14}{5} \end{aligned}$$

$\text{AROC} = \frac{14}{5} = 2.8$